NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

# Software Engineering

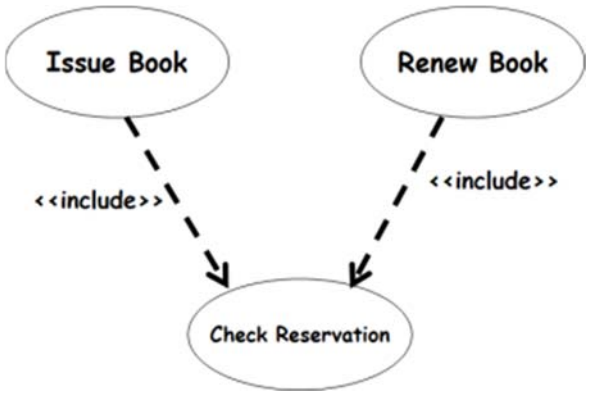
**Assignment-7**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **10** **Total mark: 10 X 1 = 10**

**QUESTION 1:**

Consider the following use case factorization example. Which of the following assertions can be inferred from it?




a. Issue book use case implicitly invokes Check reservation use case

b. Issue book use case invokes Check reservation use case only if some conditions are satisfied

c. Issue book and Renew book use cases are inter dependent

d. This factorization helps in reducing duplication of code in the implementation

e. The factorization helps in reducing the runtime complexity of the two use cases

**Correct Answer: a.** Issue book use case implicitly invokes Check reservation use case

**d.** This factorization helps in reducing duplication of code in the implementation

**Detailed Solution:**

The given diagram depicts that the issue book use case implicitly invokes the check reservation use case. By factoring out common logic (e.g., checking reservation) into a separate reusable use case, you avoid code duplication, enhancing modularity.

**QUESTION 2:**

Which one of the following are **FALSE** about a use case?

a. It is a way in which a system can be used by the users to achieve specific goals

b. It corresponds to a non-functional requirement.

c. It defines external behavior without revealing internal structure of system

d. It is a set of related usage scenarios tied together by a common goal

e. It states a future goal of the system

**Correct Answer:** b. It corresponds to a non-functional requirement.
e. It states a future goal of the system

**Detailed Solution:**

A use case describes functional requirements. A use case describes how the system behaves in response to user actions under current or designed conditions, not future goals or aspirations.

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**QUESTION 3:**

UML 1.X was enhanced with additional notational support to largely address modelling which one of the following types of applications?

a. Embedded computing applications

b. Artificial intelligence applications

c. Data science applications

d. Cloud applications

e. Blockchain applications

**Correct Answer:** a. Embedded computing applications

**Detailed Solution:**

UML 1.x initially provided general-purpose modeling capabilities for object-oriented systems. However, as real-time and embedded systems became increasingly important, UML was extended with additional notational elements. These enhancements were made to support modeling of embedded and real-time systems, which require precise control over timing, resource usage, and concurrency.

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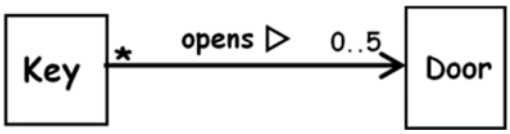
**QUESTION 4:**

Which one of the following is implied by the class diagram given below?




a. At least one key opens five doors

b. Many doors are opened by a key

c. Many doors are opened by up to 5 keys

d. A door is opened by up to 5 keys

e. A key opens up to 5 doors

**Correct Answer:** e. A key opens up to 5 doors.

**Detailed Solution:**

The given diagram implies that a key can open zero to five doors.

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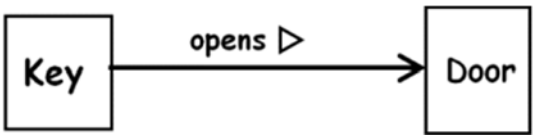
**QUESTION 5:**

Which of the following is implied by the class diagram given below?




a. Neither a Door object nor a Key object store attributes of each other

b. Either of a Door object or a Key object store attributes of the other

c. A reference of a Door object is stored as an attribute of a Key object and also a reference of a Key object is stored as an attribute of a Door object

d. A reference of a Door object is stored as an attribute of a Key object

e. A reference of a Key object is stored as an attribute of a Door object

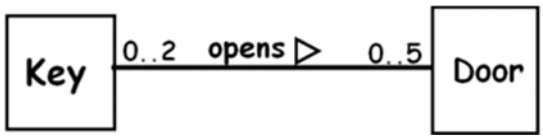
**Correct Answer:** d. A reference of a Door object is stored as an attribute of a Key object

**Detailed Solution:**

The direction of the arrow indicates where the reference is stored.

**QUESTION 6:**

Which of the following is **NOT** implied by the class diagram given below?




a. A Key opens up to 5 Doors

b. A Door is opened by up to 2 Keys

c. A Door is opened by up to 5 Keys

d. A Key may not open any of the Doors

e. Up to 2 Keys open up to 5 Doors

**Correct Answer:** c. A Door is opened by up to 5 Keys

**Detailed Solution:**

It can be observed from the given diagram that, a key can open upto 5 doors and a door can be opened by upto 2 keys.

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**QUESTION 7:**

Identification of the different categories of users in a use case diagram does **not** help towards which of the following?

a. Design of a suitable user interface

b. Design of a security and authentication mechanism

c. Design of a suitable user manual

d. Design of controller classes

e. Design of integration test plan

**Correct Answer:** d. Design of controller classes

e. Design of integration test plan

**Detailed Solution:**

Use case diagrams primarily focus on identifying users and use cases. These help in understanding who interacts with the system and what the system should do.

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**QUESTION 8:**

How is an external system represented in a use case diagram?

a. Using a specific icon available for this purpose

b. Using text description

c. Using object constraint language (OCL)

d. By stereotyping the actor icon

e. By using the symbol used to represent an object

**Correct Answer:** d. By stereotyping the actor icon

**Detailed Solution:**

An external system in a use case diagram is represented by stereotyping the actor icon.

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**QUESTION 9:**

A library automation software is to be designed. A use case named **checkReservation** is needed to check whether a book has been reserved by any user. For the **issueBook** use case, before a book can be issued it needs to be checked whether the book has been reserved by any user. Which one of the following mechanisms is most appropriate to make the **issueBook** use case to include the **checkReservation** use case?

a. Inheritance

b. Dependency

c. Include

d. Extend

e. Composition

**Correct Answer:** c. Include

**Detailed Solution:**

In Use Case Modeling, the <<include>> relationship is used when one use case always uses the behavior of another use case as part of its process.

**QUESTION 10:**

Which of the following are **NOT** implied by the class diagram given below?

a. Each course is jointly taught by two teachers

b. Each teacher teaches two courses

c. Some teachers may teach three course

d. Some teachers may not teach any course

e. Upto three Teachers can jointly teach exactly 2 Courses

**Correct Answer:** b. Each teacher teaches two courses

e. Upto three Teachers can jointly teach exactly 2 Courses

**Detailed Solution:**

The given diagram depicts that a teacher can teach upto three courses, whereas a course can be taught by two teachers jointly.

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